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An efficient Alzheimer's disease detection using optimized hybrid light GBM-based CNN

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ABSTRACT

Alzheimer's disease is a serious neurological disorder that destroys brain tissue, resulting in dementia or irreversible memory loss. Many people lose their lives to this incurable disease every year. On the other hand, early identification is essential to slowing its rapid growth. Recent studies encountered difficulties with overfitting, low accuracy, and the laborious nature of manual diagnosis. In order to resolve this, a hybrid light GBM-based CNN using Parabuteo Finch Optimization is developed for effective Alzheimer's disease identification. The combined approach of multi-level Bayesian fuzzy clustering (BFC) texture-based segmentation aims to accurately distinguish and identify specific regions corresponding to grey matter, white matter, and cerebrospinal fluid within brain MRI images. The hybrid light GBM-based DCNN provides a standardized hybridization that enables the model to harness both structured and image data. This integration enhances the capacity to capture a comprehensive set of features relevant to Alzheimer's disease detection, offering a more robust approach. The Parabuteo Finch Optimization (PFO) integration creates a versatile and efficient optimization strategy for diverse scenarios. Furthermore, the PFO algorithm enhances the overall performance and adaptability of the model by optimizing its key parameters for improved results. The research evaluates outcomes through metrics encompassing an accuracy of 94.48%, the Mathew correlation coefficient (MCC) of 0.91, the negative predictive value (NPV) of 0.92, the positive predictive value (PPV) of 0.95, and the threat score of 0.85 based on 90% of training. These results signify superior performance, establishing the effectiveness of the proposed model in comparison to alternative approaches.

1. Introduction

Alzheimer's disease (AD) is a degenerative brain disease that progressively worsens cognitive function [10]. It is the most prevalent kind of dementia, and although it often begins softly, major difficulties can arise very fast [4]. This neurodegenerative disease is distinguished by a first asymptomatic phase that lasts for roughly 20 years before symptoms appear and is marked by neuronal degeneration [11]. The subsequent early symptomatic stage, described as mild cognitive impairment owing to AD, is characterized by a reduction in cognitive function when the prodromal phase of AD is assumed to be the source of the cognitive decline [12]. Every year, 15 % of MCI patients transition to AD, making MCI a useful diagnostic tool for anticipating AD [13]. Despite the fact that there is now no known treatment for AD, it is imperative to have a diagnosis as soon as feasible to start therapies that can stop the disease's progression and improve management [14]. However, because cognitive deterioration occurs in healthy aging, diagnosing MCI can be

challenging. This necessitates using diagnostic biomarkers for AD and MCI. When making a clinical choice, one of the most commonly utilized metrics for both diagnoses is medial temporal atrophy (MTA), which can be found visually on a structural MRI scan [15]. However, MTA is not particularly useful for the early detection of MCI because it does not show until quite late in the progression of the disease [17]. Other frequently used biomarkers for MCI and AD, including fluorodeoxyglucose or amyloid positron emission tomography and brain fluid measures, are less helpful for an early, first-line diagnosis than MRIs, which are more readily available, less expensive, and less invasive [1].

Structural MRI offers valuable insights into brain structure; however, conventional diagnostic methods exhibit limitations in sensitivity and specificity when detecting Alzheimer's disease (AD). Traditional approaches struggle to capture the subtleties of early-stage AD, emphasizing the necessity for advanced technologies. The intricate structural changes in the brain that precede overt symptoms require more nuanced and precise diagnostic tools [16]. This highlights the crucial role that

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cutting-edge imaging technologies, including deep learning on structural MRI, play in improving our capacity to identify and diagnose AD in its early stages and overcoming the drawbacks of traditional approaches. MRI serves as a powerful tool offering intricate details about the anatomical integrity of the brain. This imaging modality enables the visualization of crucial aspects such as atrophy, tissue abnormalities, and structural changes linked to neurodegenerative diseases like Alzheimer's. The richness of information embedded in MRI scans, capturing both macroscopic and microscopic features, positions it as a valuable modality for in-depth analysis using deep learning [18]. The high spatial resolution and multi-dimensional data provided by structural MRI empower deep learning algorithms to discern subtle patterns and abnormalities. By taking advantage of this abundance of data, it becomes possible to identify neurodegenerative conditions precisely, early on, and accurately. This marks an evolution in medical imaging and significantly improves our understanding of and capacity for diagnosing brain diseases [19].

Deep learning's process lies in extracting hierarchical representations from intricate data, a capability exceptionally suited for unveiling nuanced patterns within medical images. This aptitude is particularly crucial in the context of Alzheimer's disease, where subtle structural alterations manifest before overt symptoms [21]. Deep learning algorithms, through their ability to discern intricate features and subtle variations, become indispensable tools for early diagnosis. The richness of information embedded in medical images, especially structural MRI scans, aligns seamlessly with deep learning's strength, promising a transformative impact on the accuracy and efficiency of AD detection. Because of this, applying deep learning to this field has a lot of potential to improve our knowledge, diagnosis, and eventually, Alzheimer's disease therapy [20]. The field of disease diagnosis and prognosis has seen a significant shift in the use of deep learning for medical imaging in recent years. This work builds on existing developments by presenting a novel method of diagnosing AD. By applying deep learning to the analysis of complex structural MRI data, the study significantly advances the quickly developing field of medicine. The integration of cutting-edge technology into Alzheimer's diagnosis reflects a commitment to advancing healthcare methodologies, offering the promise of enhanced accuracy and early detection in the quest for effective treatment strategies [22].

The ultimate aim of the research is to overcome the existing limitations, such as complex structure, overfitting issues, computational complexity, and poor generalizability, with limited resources. To effectively identify AD, a PFO-hybrid light GBM-based CNN model is developed and evaluated using the ADNI dataset. Initially, preprocessing is performed to reduce inherent noise and artifacts. A Multi-level BFC Texture-Based Segmentation technique precisely separates brain tissues. Feature extraction includes statistical features, hybrid texture patterns, and VGG-16. The strengths of Hybrid Light GBM and CNN are combined for Alzheimer's detection. The PFO algorithm optimizes the segmentation process and the hybrid light GBM model, which helps to improve the performance and address overfitting issues. The main contributions are given as follows,

Parabuteo Finch Optimization: The Parabuteo Finch Optimization method incorporated Sparrow search optimization and Harris Hawk algorithms. It combines the behavior of Sparrow explores safe environments and triggers alarms when predators are present, and Harris Hawk evaluates prey using energy and movement cues. This combination forms a flexible and effective optimization strategy that is designed for tuning the hybrid light GBM and helps in segmentation.

Hybrid Light GBM-based Deep CNN: The hybrid light GBM-based CNN is developed through the combination of light GBM and CNN, providing a standardized hybridization that enables the model to harness both structured and image data. This integration enhances the capacity to capture a comprehensive set of features relevant to Alzheimer's disease detection, offering a more robust approach.

The rest of the document is arranged as follows: [Section 2](#) analyzes

current approaches to AD detection in detail and points out their drawbacks. The proposed PFO-hybrid light GBM-based CNN's mathematical models and AD detection approach are described in detail in [Section 3](#). A comprehensive explanation of the results obtained with the PFO-hybrid light GBM-based CNN is presented in [Section 4](#). Concluding insights for this phase are detailed in [Section 5](#).

2. Literature review

The literature review of AD detection is explained as follows.

Vasco Sá Diogo et al. [1] developed a machine learning classifier that performed rather well in the multi-diagnostic classification of AD people, MCI patients, and healthy controls. While the approach showed advantages in disease differentiation and multi-diagnostic capability, it was noted to be dependent on data quality and struggled with algorithm complexity. A CNN classifier was presented by Bin Lu et al. [2], underscoring the usefulness of brain imaging as a supplemental aid in the diagnosis of AD. The protocol's remarkable generalizability was confirmed by three separate dataset validations as well as leave-site-out testing. However, the model struggled with the lack of interpretability and computational complexity in the results.

Mahda Nasrolahzadeh et al. [3] used genetic algorithms as a data-driven evolution computation-based modeling technique to create a unique system for AD diagnosis. The approach proved to be highly accurate in discriminating and appropriate for handling intricate problems. Nonetheless, it exhibited a dependency on speech signal quality and raised concerns about overfitting. A multi-label classification method for six phases of AD was described by Abdulaziz Alorf et al. [4], combining deep learning and resting-state functional MRI. While deep learning facilitated the automatic learning of complex patterns, there was a potential risk of overfitting.

A multiview-slice attention and three-dimensional CNN-based AD diagnosis model was presented by Lin Chen et al. [5]. The method considered the three-dimensional nature of brain structural magnetic resonance imaging data, offering a more comprehensive understanding of structural alterations associated with Alzheimer's disease. However, the model exhibited computational complexity and lacked interpretability. A unique deep triplet network was utilized by Maysam Orouskhani et al. [6] as a metric learning strategy for AD detection and brain MRI analysis. The approach aimed to prevent overfitting and enhance model accuracy, especially when dealing with a scarcity of image samples. Nevertheless, it faced challenges related to computational intensity and interpretability.

A transfer learning classification framework for multi-stage Alzheimer's disease detection was presented by Taher M. Ghazal et al. [7]. The method did not require hand-crafted features and demonstrated speed and efficacy in handling small image datasets. However, it encountered issues related to computational intensity and overfitting. Khalil Alsharabi et al. [8] concentrated on creating an EEG signal analysis-based AD diagnosis system. While the method enhanced the system's ability to capture various aspects of EEG signals, it was noted to have challenges related to overfitting.

Najmul Hassan et al [30] proposed a residual-based multistage deep learning model and utilized SoftMax, SVM, and RF classifiers to achieve optimal classification, which ensures robust and accurate AD classification results with multiple datasets. However, the method has computational complexity and struggles with the data imbalance issue. Turker Tuncer et al [32] presented a lightweight Fibonacci model and incorporated a CNN model with an attention mechanism. Although the FiboNeXt model with intricate block structures enhances robustness, it also generates complexity and overfitting issues.

2.1. Challenges

- The ML ensembled classifier with tree-based algorithms allowed for the development of a complex classifier performance in the multi-

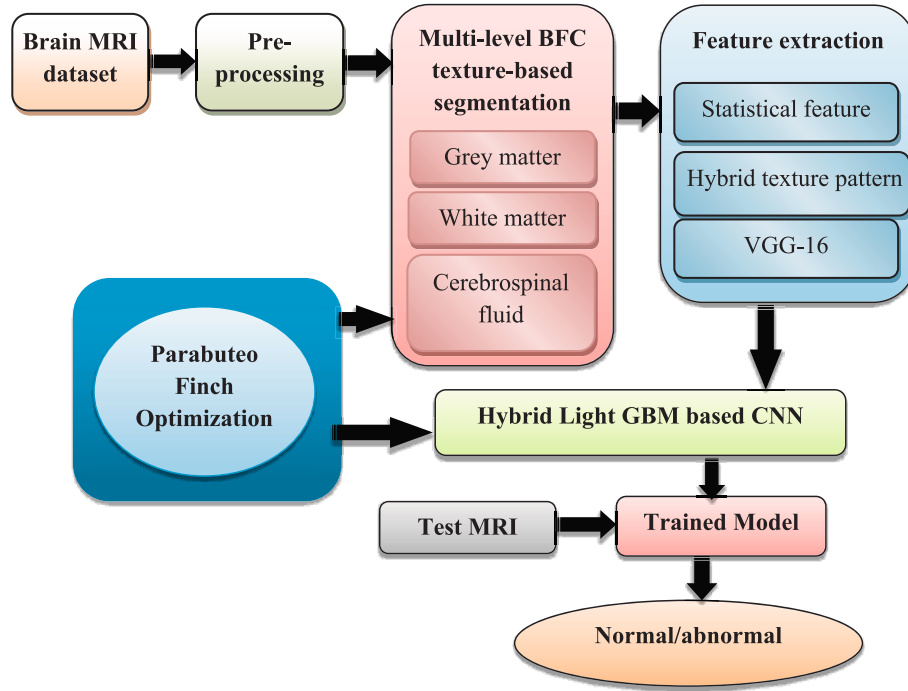


Fig. 1. Block diagram of the proposed AD detection method.

diagnosis AD detection, which also failed to recognize cognitive features and create complexity [1]

- A practical brain MRI-based AD diagnostic classifier using deep learning with Surface-based algorithms are more time-consuming during the process [2].
- The complex architecture of the Global Fourier-Net affected the model's performance in AD diagnosis and faced overfitting issues [9].
- Utilization of neuropsychological markers leads to misclassification bias in the detection. Due to complex structures and missing data, the performance was limited [10].
- AD and MCI detection using OCT-based deep-learning algorithms has poor generalizability in terms of small samples and poor applicability in real-world scenarios [11].
- Early-stage Alzheimer's disease prediction with ML models struggled to identify relevant attributes that can detect Alzheimer's disease very early [16].

3. Proposed methodology

The ultimate aim of the research is to identify AD using a PFO-hybrid light GBM-based CNN. The data is initially gathered from the ADNI dataset [23], which comprises a collection of MRI images. These dataset images contain inherent noise and artifacts, so preprocessing is performed to reduce noise, ensuring a cleaner and more accurate representation of brain structures. Subsequently, a Multi-level Bayesian Fuzzy Clustering Texture-Based Segmentation method is utilized to accurately distinguish between the cerebrospinal fluid, white matter, and grey matter in the brain MRI images. This segmentation process is further optimized using Parabuteo Finch Optimization, enhancing the accuracy of tissue classification. After that, feature extraction is performed through three steps, such as statistical features, hybrid texture pattern and VGG-16. A thorough representation of the various brain areas is enhanced by the features that were extracted. Next, a CNN based on Hybrid Light GBM is developed to identify Alzheimer's disease. This model combines the strengths of Light GBM and CNN to provide a robust classification framework. The model is further optimized using Parabuteo Finch Optimization algorithms to fine-tune parameters. Lastly, the

trained model is assessed on MRI slices to differentiate between normal and aberrant cases in order to determine whether AD is present. Fig. 1 shows the block diagram of the proposed approach.

3.1. Input

The data is gathered from the ADNI dataset [23], which includes a collection of MRI images and is expressed mathematically as,

$$A = \sum_{i=1}^p A_n \quad (1)$$

where, A represents the dataset, A_n represents the images in the dataset, varies from 1 to p

3.2. Pre-processing

The MRI brain image contains noise from electronic image sensors, sensor temperature, insufficient light, film grain, and channel noise. The preprocessing technique is performed in order to make the input image sharper and to remove blur. Various artifacts, such as susceptibility artifacts or radiofrequency artifacts, can affect MRI images. The main goals of preprocessing are to enhance image quality, correct for artifacts, and standardize the data for more accurate and meaningful analysis. The mathematical equation becomes,

$$A^* = \sum_{i=1}^p A_n^* \quad (2)$$

where, A^* denotes the pre-processed dataset.

3.3. Multi-level BFC texture-based segmentation

Multi-level Bayesian Fuzzy Clustering (BFC) texture-based segmentation for MRI images is an image analysis approach that combines Bayesian principles, fuzzy clustering, and texture features to partition magnetic resonance imaging (MRI) data into distinct regions. This method considers uncertainties in the data, incorporates multiple levels

of information, and focuses on capturing complex texture patterns specific to MRI. Clustering is a technique used in image segmentation to group pixels with similar properties. A data point might have various degrees of membership in multiple groups through the use of fuzzy clustering. This is especially helpful for complicated images where it may be difficult to distinguish the boundaries between several tissues. Texture in image processing is the way that pixel intensities are arranged in space. Different textures may be seen in different tissues or structures within an image. Texture-based segmentation approach involves using the texture patterns within an image to delineate and segment different regions. Further, the applicability of the PFO algorithm boosts a more accurate segmentation process. In the case of brain MRI, grey matter, white matter, and cerebrospinal fluid may exhibit different textures. This method's main objective is to accurately identify and distinguish the regions of brain MRI scans that correspond to white matter, grey matter, and cerebrospinal fluid. Each of these tissues may have distinct textures and intensity patterns that can be exploited for accurate segmentation. The detailed explanation of the BFC model is in the section below.

Bayesian Fuzzy clustering model:

In BFC, input data points are described as Gaussian random variables, while membership vectors are expressed as Dirichlet-distributed random variables. Since Gaussian distributions are members of the exponential family, convex combinations can be used to indicate the natural parameters of the Gaussian that represents each data point. Weighting the natural parameters of each cluster with the corresponding partial membership values creates these combinations. When estimating the Bayesian model, one must take into consideration the total potential of the information points and elements that comprise the Bayesian model as a whole. This estimation process encompasses the integration of the Dirichlet distribution for membership vectors and Gaussian distributions for data points, capturing the underlying uncertainty and relationships within the data through a probabilistic framework.

$$q(E_j^i, V^*, B^*) = p(E_j^i, V^*, B^*) q(B^*) \tilde{q}(V^* | B^*) \quad (3)$$

where, $q(B^*)$ represents the Gaussian prior distribution (GPD), $\tilde{q}(V^* | B^*)$ refers to the fuzzy cluster prior, and $p(E_j^i, V^*, B^*)$ indicates the fuzzy data likelihood distribution. The following FDL-based formula can be used to obtain the BFC model.

$$q(E_j^i, V^*, B^*) = \prod_{m=1}^n FDL(e_m | b_m, B) \quad (4)$$

where V^* stands for the fuzzy relationship regulator, E_j^i for the information matrix, and B^* for the cluster prototypes. n represents the total number of information points, while e_m represents the number of data points in a cluster B . The computation of B^* is linked to the likelihood of the data, and a traditional probability is called FDL. The clusters are shaped in a manner where they collectively share mean values, and these shared mean values define the cluster models.

$$q(E_j^i, V^*, B^*) = \prod_{m=1}^n \frac{1}{(Y(B_m, n, B))} \prod_{l=1}^F M(e_m | \gamma = a_l, d = b_{ml} J) \quad (5)$$

where, $Y(B_m, n, B)$ represents the normalization constants, B the cluster, F the total number of clusters, l the cluster number, and n the fuzzifier. FCP mimics the performance of FCM by applying the Bayesian model. It may be expressed as,

$$\tilde{q}(V^* | B^*) = \prod_{m=1}^n FCP(b_n | B^*) \quad (6)$$

There are three components in the FCP, including $Y(B_m, n, B)$ $\prod_{l=1}^F c_{ml}^{-nk/2}$ and Dirichlet $(b_m | \gamma)$.

$$\tilde{q}(V^* | B^*) = \prod_{m=1}^n Y(B_m, n, B) \left(\prod_{l=1}^F c_{ml}^{-nk/2} \right) Dir(b_m | \gamma) \quad (7)$$

where γ indicates the mean factor and B_{ml} indicates the data point's membership in group l .

In order to improve the stability of the FDL, the first factor removes the FDL stabilization constant. The subsequent influence is geared towards yielding elevated membership estimates. The third factor introduces enthusiasm into the process, contributing added flexibility and efficiency to the clustering procedure. The Generalized Possibilistic Distribution (GPD) is expressed as follows,

$$q(B^*) = \prod_{l=1}^F M \left(a_l | \gamma_a, \sum_a \right) \quad (8)$$

The dataset mean is denoted by γ_a

$$\gamma_a = \frac{1}{n} \sum_{m=1}^n e_m \quad (9)$$

The data covariance is represented by $\sum_a$

$$\sum_a = \frac{\eta}{n} \sum_{m=1}^n (e_m - \gamma_a)^* (e_m - \gamma_a)^T \quad (10)$$

where, η stands for the user-imposed restriction that tampers with the strong point. Using these components, the joint probability of the parameters and data is created.

3.4. Feature extraction

After segmentation, feature extraction is performed through three steps, such as statistical features, hybrid texture pattern and VGG-16. The goal of feature extraction is to create a comprehensive and discriminative representation of the brain regions. By combining statistical features, hybrid texture patterns, and features extracted from a deep neural network like VGG-16, the extracted features aim to capture a broad range of information, from basic statistical characteristics to complex hierarchical patterns present in the segmented regions. The extracted features serve as input to subsequent stages of the analysis, such as classification or further refinement of the segmentation results. The detailed explanation of features are in the section below.

3.4.1. Statistical features

Statistical features encompass the calculation of diverse statistical measures or descriptors derived from the pixel values within a specified image region. These features comprise metrics such as mean, variance, skewness, kurtosis, and other moments characterizing the distribution of pixel intensities. By quantifying the pixel intensity distribution, these statistical features offer numerical insights into the shape, intensity variability, and overall characteristics of the region, providing a comprehensive understanding of its properties.

i) Mean: The mean, serving as a measure of central tendency, signifies the average intensity value across all pixels within a defined ROI in an MRI image. Its computation involves summing up the pixel values within the specified region and dividing the total by the number of pixels, providing a representative indicator of the overall intensity level in that particular region.

$$\mu = \frac{\sum y}{H} \quad (11)$$

where y is the total of all data values, H is the number of data points, and μ is the mean.

ii) Variance: Variance serves as a metric for the spread or dispersion of pixel intensity values within a given region, reflecting the extent of deviation from the mean. This measure quantifies how much individual

pixel values differ from the average. The process computes the average of the squared deviations between the mean and each pixel value, yielding a measure of the overall intensity variability in the designated area.

$$\sigma^2 = \frac{\sum (y - \mu)^2}{H} \quad (12)$$

where σ^2 is the variance, $(y - \mu)$ represents the variation among each data point and the mean.

iii) Standard deviation: The standard deviation serves as an additional gauge of the dispersion of pixel values within a specified region. As the square root of the variance, it offers a more easily interpretable measure of the spread. The calculation of the standard deviation involves taking the square root of the variance, providing a concise and accessible assessment of the overall variability in pixel intensity within the defined region.

$$\sigma = \sqrt{\sigma^2} \quad (13)$$

3.4.2. Hybrid texture pattern

The hybrid texture pattern probably describes a synthesis of various models or techniques for texture analysis. The spatial organization of pixel intensities is captured by texture patterns, which are useful for differentiating between various tissues or structures in an image. The ability to capture various textural features inside different brain regions is improved by combining numerous texture patterns or approaches, which may lead to an improvement in the accuracy of feature representation. The next section provides a full discussion of LDP, LBP, and LDP characteristics.

i) Local Binary Pattern (LBP): Pixel intensity patterns in an image's local patterns are encoded using a texture descriptor called LBP. Each pixel's intensity is compared to that of its nearby pixels to determine how it operates, and the outcome is shown as a binary code. One can generate a binary code for each pixel in the image by comparing its nearby intensities to the center pixel. This allows one to identify whether adjacent pixels are darker or brighter than the center pixel.

$$LBP_{K,S} = \sum_{k=1}^K 2^{(k-1)} \times f_1(r_k - r_c) \quad (14)$$

$$f_1(z) = \begin{cases} 1 & z \geq 0 \\ 0 & \text{else} \end{cases} \quad (15)$$

where S is the radius of the neighborhood, K is the number of neighbors, and r_k is the gray value of the adjacent pixels of the center pixel.

ii) Local Ternary patterns (LTP): Pixel intensities are compared over three intensity levels rather than simply two in LTP, an expansion of LBP. Brightness, darkness, and resemblance are the three characteristics that are used to categorize the relationship between the main pixel and its neighbors. Like LBP, LTP uses the relationships between the center of a pixel and its neighbors to generate a ternary code for every pixel. LTP works well in situations where minute texture fluctuations are crucial since it is made to capture more complex local intensity changes in images.

$$f_1(z, r_h, u) = \begin{cases} \pm 1 & z \geq r_h + u \\ 0 & |z - r_h| < u \\ 1 & z \leq r_h - u \end{cases} \quad (16)$$

iii) Local derivative patterns (LDP): Directional information is integrated into the analysis using LDP as a texture descriptor. When comparing LDP to other approaches, it must take into consideration not just the spatial orientation of the neighboring pixels but also the intensity connections between the central pixel and them. Through the use of directional patterns, LDP encodes important details about how different directions of intensity changes are arranged around each pixel,

adding to a more complete depiction of texture properties.

$$LDP_{\beta}^w(r_h) = \sum_{k=1}^K 2^{(k-1)} \times f_2(I_{\beta}^{(w-1)}(r_h), I_{\beta}^{(w-1)}(r_k)) \Big|_{k=8} \quad (17)$$

$$f_2(z, y) = \begin{cases} 1 & \text{if } z, y \leq 0 \\ 0 & \text{else} \end{cases} \quad (18)$$

3.4.3. VGG-16 features

The CNN architecture known as VGG-16 consists of 16 wt layers, which include convolutional and fully connected layers. Originally designed for image classification tasks, this model leverages its 16 layers to extract intricate features from input images. When employing a pre-trained deep learning model like VGG-16 for feature extraction, the focus is on utilizing the learned representations from various layers of the network. Deep neural networks, and VGG-16 in particular, extract hierarchical features from input images that allow for the extraction of complex patterns and representations. Specifically, VGG-16 streamlines the design by emphasizing ConvNet layers with a 3×3 kernel size. Its importance arises from the fact that pre-trained values are readily accessible and integrated into systems and applications through the internet. Compared to other comprehensive models, VGG-16 stands out for its simplicity, making it accessible for a variety of applications. The model can accommodate typical image dimensions; its minimum required input image size is 224×224 pixels with three channels.

3.5. Hybrid light GBM-based DCNN

The hybrid light GBM-based DCNN is developed by the hybridization of light GBM and DCNN. Light GBM might be employed to handle structured data associated with Alzheimer's disease detection, such as patient demographics, clinical history, or genetic information. DCNNs excel at extracting spatial features from images, making them well-suited for identifying patterns indicative of Alzheimer related brain changes. Integrating Light GBM with DCNNs allows the model to leverage both structured and image data, potentially capturing a more comprehensive set of features relevant to Alzheimer's disease detection. By combining the advantages of both Light GBM and DCNN, the model is able to recognize intricate patterns and relationships within the data. This could result in more precise and trustworthy predictions for the early identification of AD. Light GBM's ability to handle missing values in structured data, combined with DCNN's ability to focus on relevant image regions, creates a more robust model capable of handling incomplete patient information. The hybrid model leverages the strengths of each component to utilize diverse data sources effectively. The detailed explanation of light GBM and DCNN is discussed in the section below.

3.5.1. Light gradient boosting machine (GBM)

Light GBM stands out as a gradient boosting framework crafted for efficiency and scalability, particularly well-suited for handling large datasets. Belonging to the family of decision tree-based ensemble learning methods, Light GBM is renowned for its speed and memory efficiency. Its unique approach involves constructing decision trees in a parallel and leaf-wise fashion, providing notable advantages in terms of computational efficiency. In the context of a training set represented as $\{(i_1, j_1), (i_2, j_2), G, (i_v, j_v)\}$, where i denotes data samples and j signifies class labels, let $E(i)$ symbolize the expected function. The Gradient Boosting Decision Tree (GBDT) optimization goal is to reduce the loss function $G(j, E(i))$:

$$\hat{E} = \underset{E}{\operatorname{argmin}} F_{ij}[G(j, E(i))] \quad (19)$$

Next, use a line search to reduce the loss function to achieve the GBDT iterative criterion.

$$E_n(i) = E_{n-1}(i) + \phi_n t_n(i) \quad (20)$$

where $t_n(i)$ is the base decision tree, $\phi_n = \arg\min_{\phi} \sum_{x=1}^N G(i_x, E_{n-1}(i_x) + \phi t_n(i_x))$, and n is the number of iterations.

However, the effectiveness and accuracy of GBDT may not produce results that are sufficient when working with an extensive number of samples or an excessive number of feature dimensions. Finding the ideal split points throughout the tree learning process is a computationally intensive task for GBDT, an ensemble technique that uses decision trees as base classifiers. Information gain is frequently utilized in GBDT to divide each node. In contrast, Light GBM uses variance gain to calculate split points using the Gradient-based One-Side Sampling (GOSS) approach. First, the absolute values of the slopes for the training instances must be sorted in descending order. The top $h \times 100\%$ data samples are then chosen and designated as C based on gradient values. Next, a random selection is made from the retained samples C^g to create the subset D , whose size is $d \times |C^g|$. The instances are finally divided by the expected $L_y(p)$ on $C \cup D$.

$$L_y(p) = \frac{1}{w} \left\{ \frac{\left(\sum_{i_x \in C_l} o_x + \frac{1-h}{c} \sum_{i_x \in D_l} o_x \right)^2}{v_l^y(p)} + \frac{\left(\sum_{i_x \in C_u} o_x + \frac{1-h}{c} \sum_{i_x \in D_u} o_x \right)^2}{v_u^y(p)} \right\} \quad (21)$$

where, $C_l = \{i_x \in C : i_{xy} \leq p\}$, $C_u = \{i_x \in C : i_{xy} > p\}$, $D_l = \{i_x \in D : i_{xy} \leq p\}$, $D_u = \{i_x \in D : i_{xy} > p\}$, o denotes the loss function's negative gradient and $\frac{1-h}{c}$ is used to normalize the sum of gradients.

High-dimensional datasets frequently contain sparse characteristics, many of which are exclusive. The training of GBDT can be sped up by using Efficient Feature Bundling (EFB). To be more precise, Light GBM can combine unique features into a single feature bundle, from which an analogous feature histogram can be created using a feature scanning technique. This method helps to maximize training efficiency, especially when working with sparse and high-dimensional datasets. Consequently, from $Q(\text{data} \times \text{bundle})$ to $Q(\text{data} \times \text{feature})$ the computational complexity of Light GBM is lowered, where $\text{bundle} \times \text{feature}$.

With GOSS and EFB, Light GBM is a useful GBDT implementation that increases computing efficiency without sacrificing accuracy. Variance gain is calculated, and then the best node is split using GOSS. By fusing a large number of unique characteristics with less dense features, EFB helps speed up GBDT training. Decision trees are used by the ensemble model Light GBM as its foundation classifier. By fitting the negative gradients of the loss function, these trees can be trained successively. Equation (20) states that the weighted combination approach can be used to produce the Light GBM model.

$$E_N(i) = \sum_{n=1}^N \phi_n t_n(i) \quad (22)$$

where $t_n(i)$ is the initial decision tree and N is the greatest number of iterations.

3.5.2. Deep CNN classifier

The CNN is an advanced neural network structure that has evolved from the traditional multilayer perceptron method. It comprises two main components: the first part, known as the convolutive part, acts as an image feature extractor. The image undergoes a series of filtering processes, generating new images known as convolution maps. Some intermediate filters employ a local maximum operation to reduce the image resolution. These convolution maps are then joined to create a feature vector. This feature vector is then fed into the second part, which is the classification component consisting of fully connected layers, akin to a multilayer perceptron. These layers enable the classification of

images into two classes. The network architecture involves 5 convolution layers, each employing 3×3 filters with 32 filters per layer. Each convolution layer is associated with a pooling layer. The final output is obtained through a fully connected layer, completing the classification process.

The convolution layer operates by moving a filter systematically across the entire image and performing a convolution (matrix product) between this filter and the underlying image. Initially, the filter is positioned at the top left of the image, and it shifts a specified number of boxes (the step size) to the right. Upon reaching the image's end, the filter shifts one step down and repeats the process until it has traversed the entire image. This method aims to highlight specific characteristics of the given image. The pooling layer serves the purpose of reducing the image size while retaining its pertinent characteristics. A commonly used method is Max Pooling, which involves downsizing the image while preserving the highest pixel values. At each position of the tile, the maximum pixel value is extracted. This results in a new image that retains only the salient features of the original image.

The correction layer, often referred to as the Rectified Linear Unit (ReLU) layer, is incorporated to enhance processing efficiency. Placed between treatment layers, this layer applies an activation function to the output signals, enforcing neurons to produce positive values. The flattening operation involves converting all image matrices into a single vector. Next, using this vector, the first fully connected neuron layer is created, with each value coupled to a neuron in the first network layer. Following multiple convolution and max-pooling layers, the classification process is executed through fully connected layers. Neurons in this layer establish connections with all neurons in the preceding layer. To obtain probability distribution vectors, the outputs of this layer undergo processing by the Softmax function. A vector of size M , which is the number of classes in this image classification issue, is produced by this function. Every component of the vector represents the likelihood that the supplied image falls into a particular class.

3.6. Proposed Parabuteo finch optimization

The Parabuteo Finch Optimization is a novel approach that emerges from the fusion of the Sparrow Search Algorithm [28] and the Harris Hawk Optimization Algorithm [29]. By integrating the collaborative hunting strategy of Harris Hawk and the exploring, alerting behavior of Sparrow creates a *meta*-heuristic algorithm called Parabuteo Finch Optimization algorithm is created, offering a more effective and versatile optimization strategy to find optimal solutions. The PFO algorithm fine-tunes the hyperparameters of the model for accurate AD detection with fast convergence.

Motivation

The Hunting Behavior of Harris Hawks (HHO) is known for their collaborative hunting strategy, where they work together in a coordinated manner to capture prey. The motivation behind developing Harris Hawk Optimization lies in emulating the efficiency and effectiveness of this natural behavior for problem-solving in optimization tasks and improving the algorithm's capacity for exploration and exploitation. Finding a balance between exploration and exploitation is the goal of Harris Hawk Optimization. However, the HHO algorithm easily converges to a local optimal solution. The Sparrow Search Algorithm (SSA) utilizes the foraging behavior of sparrows in their quest for food. Sparrows, being opportunistic and intelligent feeders, exhibit a range of feeding techniques, adapting their methods based on the prevailing conditions of their habitat and the nature of their prey. In such situations, one or more individuals emit a chirp, signaling the entire group to take flight. The SSA optimization provides a fast convergence rate and reduces suboptimal problems. Hence, the SSA and HHO algorithms are combined to generate Parabuteo finch optimization to enhance the global search in a complex environment and provide fast convergence.

Mathematical equation of the proposed Parabuteo finch optimization

The mathematical equation of the proposed Parabuteo finch optimization is discussed as follows,

i) Initialization:

Initially, the position of the searcher in the random population is written as,

$$X^t = X_{rand} + V_o(u - l) \quad (23)$$

where, the random position is denoted as X_{rand} , the initial velocity is denoted as V_o , the upper and lower bounds is denoted as u and l .

ii) Evaluate fitness function

The fitness of solutions is evaluated by applying the objective function to all solutions when $t < t_{max}$. In the context of fine-tuning a light GBM-based deep DCNN, the optimal solution is determined by its ability to achieve the highest accuracy in disease detection.

$$fitness\ function = accuracy \quad (24)$$

iii) Exploration and exploitation phase

Initially, the exploration and exploitation depend upon the predator. The primary goal of exploration is to ensure a broad coverage of the solution space, identifying diverse possibilities that may contain better solutions. The main goal of exploitation is to refine and improve the existing solutions, leveraging the information gained during exploration to converge towards the optimal or near-optimal solutions.

Case i) Probation phase

When $R < ST$ that is alarm value ranging from 0 to 1 ($Re(0, 1)$) falls below the safety threshold ranges from 0.5 to 1 ($STe(0.5, 1)$), indicating the absence of predators, the system enters the probation phase (exploration phase). In this scenario, the prey autonomously explores to fulfill its needs. The alarm value range from 0 to 1 underscores the absence of immediate threats.

For Sparrow, the equation becomes,

$$X^{t+1} = X^t e^{\frac{-t}{\alpha t_{max}}} + X^{t-1} e^{\frac{-(t-1)}{(1-\alpha)t_{max}}} \quad (25)$$

where α denotes the frequency of the alarming signal, t denotes the iteration and $t-1$ denotes the previous iteration.

For the Harris hawk, the equation becomes,

$$X^{t+1} = X_t - r_1 |X_{rand}(t) - 2r_2 X(t)| \quad (26)$$

where, $r_1 = \frac{[t-1]}{t_{max}} \varepsilon(0, 1)$, $r_2 = \frac{u-l}{t} \varepsilon(0, 1)$

$$X_t = X^{t+1} + r_1 |X_{rand}(t) - 2r_2 X(t)| \quad (27)$$

Substitute equation (27) in 25 then,

$$X^{t+1} = (X^{t+1} + r_1 |X_{rand}(t) - 2r_2 X(t)|) \left(e^{\frac{-t}{\alpha t_{max}}} + X^{t-1} e^{\frac{-(t-1)}{(1-\alpha)t_{max}}} \right) \quad (28)$$

$$X^{t+1} = X^{t+1} e^{\frac{-t}{\alpha t_{max}}} + (r_1 X_{rand}(t) - 2r_2 X(t)) * e^{\frac{-t}{\alpha t_{max}}} + X^{t-1} e^{\frac{-(t-1)}{(1-\alpha)t_{max}}} \quad (29)$$

$$X^{t+1} = \frac{1}{1 - e^{\frac{-t}{\alpha t_{max}}}} \left[r_1 X_{rand}(t) - 2r_2 X(t) \cdot e^{\frac{-t}{\alpha t_{max}}} + X^{t-1} e^{\frac{-(t-1)}{(1-\alpha)t_{max}}} \right] \quad (30)$$

Case ii) Prevention phase: When $R > ST$, that is alarm value greater than the safety threshold, indicating the predators are nearby, the system enters the prevention phase (exploitation phase). It aims to intensively search around the currently known optimal solutions or areas with high potential for finding the best solution.

a) Exertion phase: When the energy of the prey is greater than the threshold ($|E| \geq \gamma$) and the escaping chances of the prey is less ($r \geq \gamma$) it is called the exertion phase. In this phase, the prey has enough energy but has a low chance of escape due to the strength of the predator in count and energy.

Sparrow equation becomes,

$$X^{t+1} = X_{best}^t + \beta |X^t - X_{best}| \quad (31)$$

Harris equation becomes,

$$X^{t+1} = X_p - X_t - E_L |k_1 X_p - X_t| \quad (32)$$

where, the energy loss is denoted as $E_L = |F[X_p] - F[X_t]|/F[X_t]$, the angular constant is denoted as $k_1 = 2(1 - r_3)$, $r_3 = \cos\theta$, the angle of attack is denoted as θ .

$$X^{t+1} - X_t = X_p - E_L |k_1 X_p - X_t| \quad (33)$$

Apply fractional calculus theory,

$$D[X^{t+1} - X_t] = X_p - E_L |k_1 X_p - X_t| \quad (34)$$

$$X^{t+1} - \alpha X^t - \frac{1}{2} X^{t-1} - \frac{1}{6} (1 - \alpha) X^{t-2} = X_p - E_L |k_1 X_p - X_t| \quad (35)$$

$$X^{t+1} = \alpha X^t + \frac{1}{2} X^{t-1} + \frac{1}{6} (1 - \alpha) X^{t-2} + X_p - E_L |k_1 X_p - X_t| \quad (36)$$

Now, $X^{t+1} = \frac{1}{2}$ in equation (31) + $\frac{1}{2}$ in equation (36),

$$X^{t+1} = \frac{1}{2} [X_{best}^t + \beta |X^t - X_{best}|] + \frac{1}{2} \left[\alpha X^t + \frac{1}{2} X^{t-1} + \frac{1}{6} (1 - \alpha) X^{t-2} + X_p - E_L |k_1 X_p - X_t| \right] \quad (37)$$

$$X^{t+1} = \frac{1}{2} \left[X_{best}^t + \beta |X^t - X_{best}| + \alpha X^t + \frac{1}{2} X^{t-1} + \frac{1}{6} (1 - \alpha) X^{t-2} + X_p - E_L |k_1 X_p - X_t| \right] \quad (38)$$

b) Cooperative working phase: If $|E| < \gamma$, $r \geq \gamma$ then it enters into the cooperative working phase. In this phase, the prey has low energy, and the predator remains as a group. The prey tries to escape, but the predator diverts the hunt and successfully attacks the prey.

The Harris hawk equation becomes,

$$X^{t+1} = X_p^t - E_L |X_p - X_t| \quad (39)$$

where, $E = \left| \frac{[F(X_p) - F(X_t)]}{[F(X_t)]} \right|$

Sparrow equation becomes,

$$X^{t+1} = X^t + \left[\frac{t - u + l}{t} \right] \cdot X^{t-1} (u - l) \quad (40)$$

Now $X^{t+1} = \frac{1}{2} X_{eagle}^{t+1} + \frac{1}{2} X_{sparrow}^{t+1}$

$$X^{t+1} = \frac{1}{2} [X_p^t - E_L |X_p - X_t|] + \frac{1}{2} \left[X^t + \left[\frac{t - u + l}{t} \right] \cdot X^{t-1} (u - l) \right] \quad (41)$$

c) Intelligence phase: if $|E| > \gamma$, $r < \gamma$ it then enters into intelligence phase and in this phase, the prey has good energy and escapes from attacks due to its strength and experience in hunting. The knowledge of prey in hunting is a significant factor.

Sparrow equation becomes,

$$X^{t+1} = X^t + k_2 \frac{|X^t - X_{worst}|}{[(F(X)) - (F(X_{worst})) + \varepsilon]} \quad (42)$$

where, the experience value of prey is denoted as $\varepsilon = (1 - \rho)\varepsilon^{t-1} + \rho|E^{t-1} - E^{t-2}|$, the memory matrix is denoted as ρ .

Harris hawk equation becomes,

$$X^{t+1} = X_p(t) - E |k X_p(t) - X_t| + S \oplus \text{levy}(\beta) \quad (43)$$

where, the entry-wise multiplication is denoted as $\oplus$, the randomization

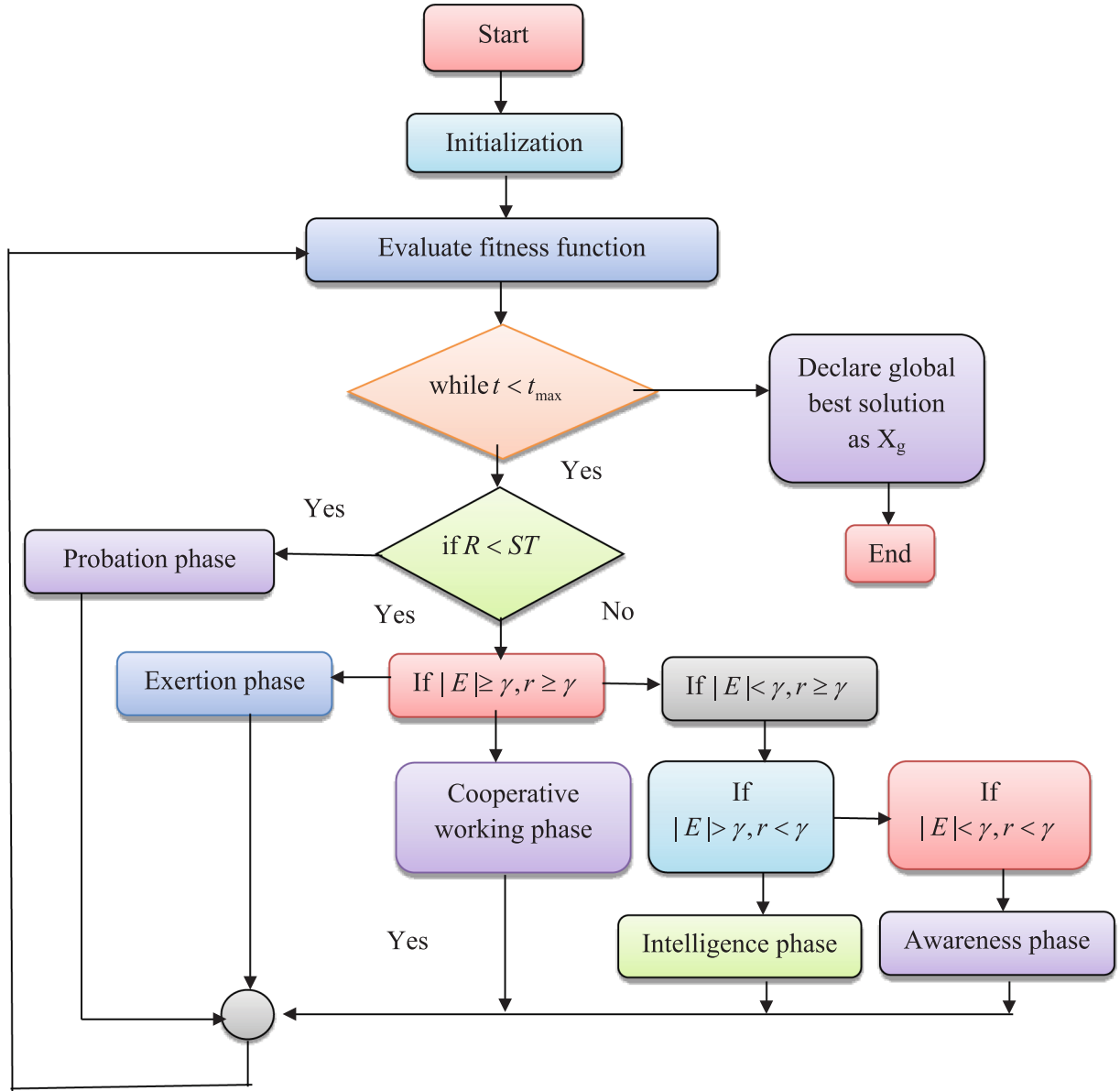


Fig. 2. Flow chart of proposed Parabuteo finch optimization.

term is denoted as S ranges from (1, 2).

$$\text{Now } X^{t+1} = \frac{1}{2} \left(X_{sparrow}^{t+1} \right) + \frac{1}{2} \left(X_{eagle}^{t+1} \right)$$

$$X^{t+1} = \frac{1}{2} \left[X^t + k_2 \frac{|X^t - X_{worst}|}{[(F(X)) - (F(X_{worst})) + \varepsilon]} \right] + \frac{1}{2} [X_p(t) - E|kX_p(t) - X_t| + S \oplus \text{levy}(\beta)] \quad (44)$$

$$X^{t+1} = \frac{1}{2} \left[X^t + k_2 \frac{|X^t - X_{worst}|}{[(F(X)) - (F(X_{worst})) + \varepsilon]} + X_p(t) - E|kX_p(t) - X_t| + S \oplus \text{levy}(\beta) \right] \quad (45)$$

d) Awareness phase: if $|E| < \gamma, r < \gamma$ it then enters into the awareness phase, and in this stage the prey has no energy, but still, it has a chance of escape due to the poor capability of the predator (having poor knowledge or it may be a young one) also the prey works as a group to escape from the predators.

For sparrow,

$$X^{t+1} = X^t + k_2 \frac{|X^t - X_{worst}|}{[(F(X)) - (F(X_{worst})) + \varepsilon]} \quad (46)$$

For eagle,

$$X^{t+1} = X_p^t - E|kX_p^t - X_{mean}^{(t)}| \quad (47)$$

where, $X_{mean}^{(t)} = \frac{\sum_{i=1}^N X_i(t)}{N}$ Here the total number of search agents is denoted as N .

While combining (46) and (47) as,

$$X^{t+1} = \frac{1}{2} \left[X^t + k_2 \frac{|X^t - X_{worst}|}{[(F(X)) - (F(X_{worst})) + \varepsilon]} \right] + \frac{1}{2} [X_p^t - E|kX_p^t - X_{mean}^{(t)}|] \quad (48)$$

$$X^{t+1} = \frac{1}{2} \left[X^t + k_2 \frac{|X^t - X_{worst}|}{[(F(X)) - (F(X_{worst})) + \varepsilon]} + X_p^t - E|kX_p^t - X_{mean}^{(t)}| \right] \quad (49)$$

Termination:

If the case $t < t_{max}$ is not satisfied, then the PFO algorithm stops its

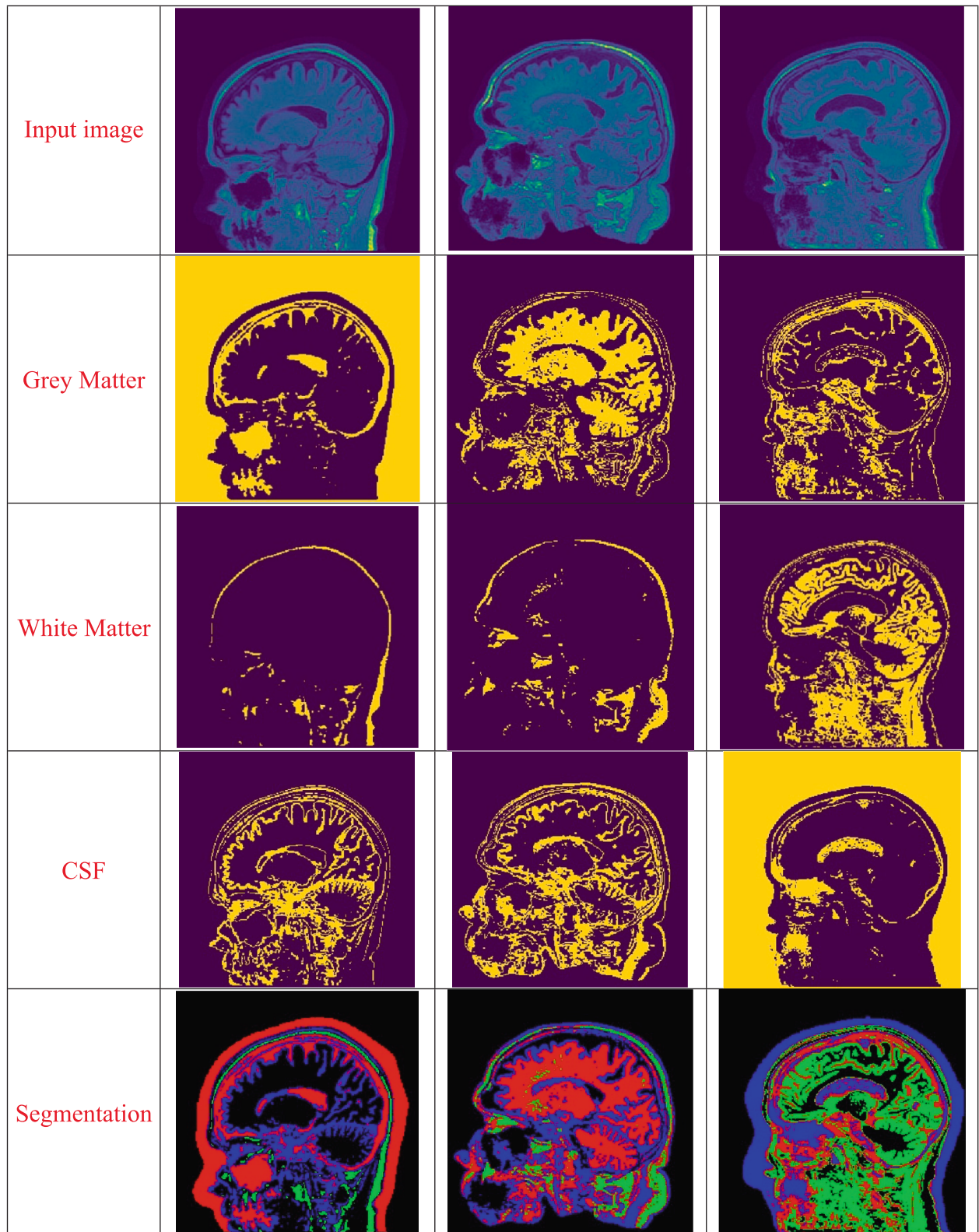


Fig. 3. Experimental results of PFO-hybrid light GBM-based deep CNN for AD detection.

iteration and delivers the optimal solution for the accurate AD detection. Thus, the exploration and exploitation phase of the sparrow and Harris hawk optimization enhances the PFO algorithm, thereby improving the ability of the proposed PFO-hybrid light GBM-based DCNN model to accurately detect the presence of Alzheimer's in MRI. The flowchart of the PFO algorithm is unveiled in Fig. 2.

4. Result

This section presents an experimental verification of the PFO-hybrid light GBM-based DCNN's efficacy for AD detection through performance assessments. Different comparison methodologies are used, and the results are evaluated according to the size of the epoch.

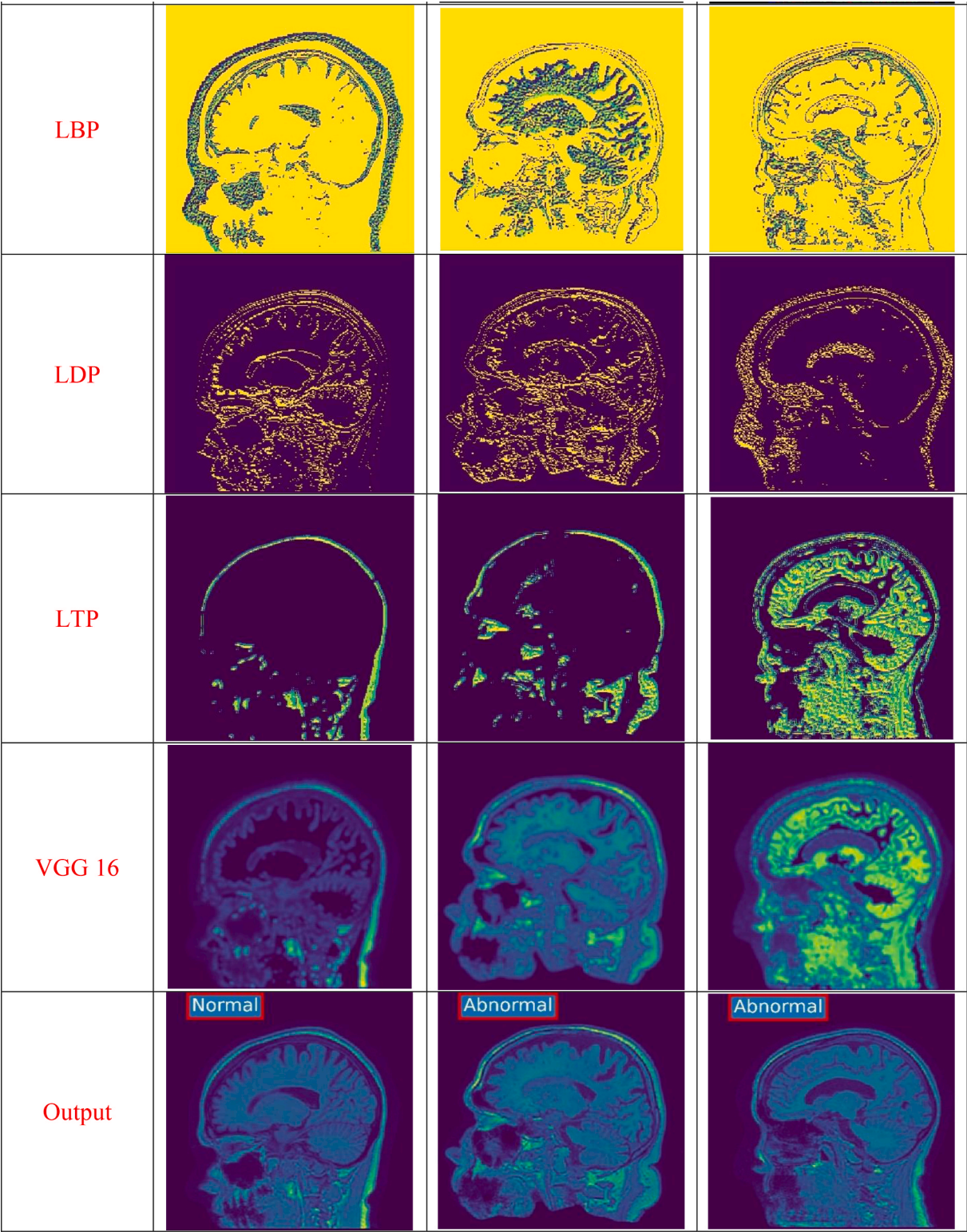


Fig. 3. (continued).

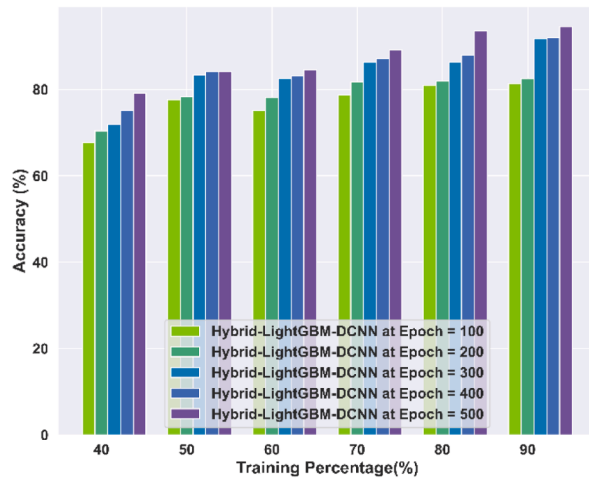
4.1. Experimental setup

The performance of the PFO-hybrid light GBM-based deep CNN classifier model is evaluated using the Python tool, and assessments comparing different approaches are conducted on a Windows 11 computer with 16 GB RAM.

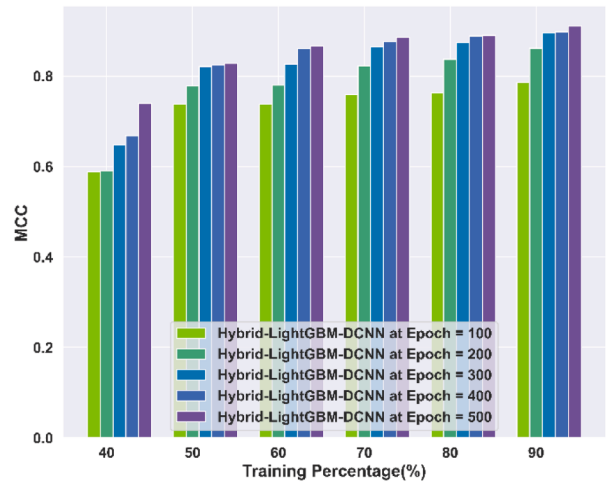
4.2. Dataset description

4.2.1. ADNI dataset [23]

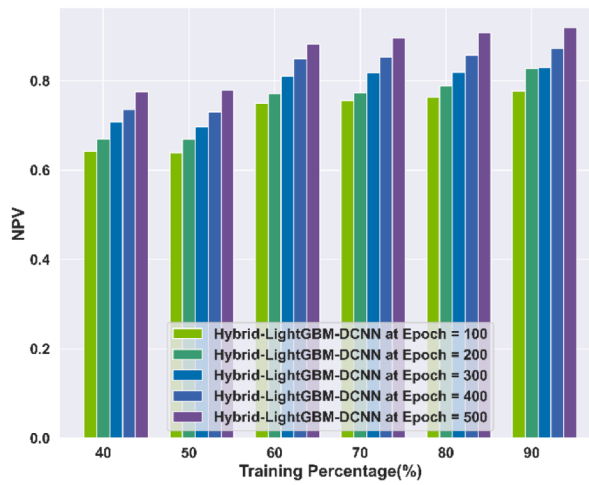
The ADNI dataset contains a wide variety of data, including clinical facts, imaging scans, biomarkers, and genetic information. This dataset, which primarily consists of 2D axial images, is extracted from the ADNI base dataset, which also includes Nifti images. The ADNI database



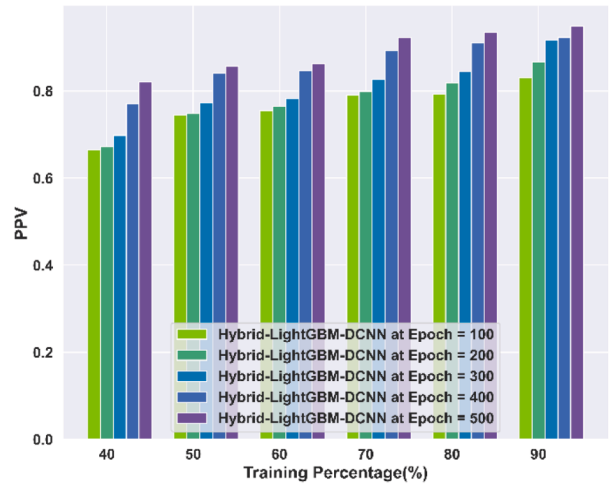
Accuracy



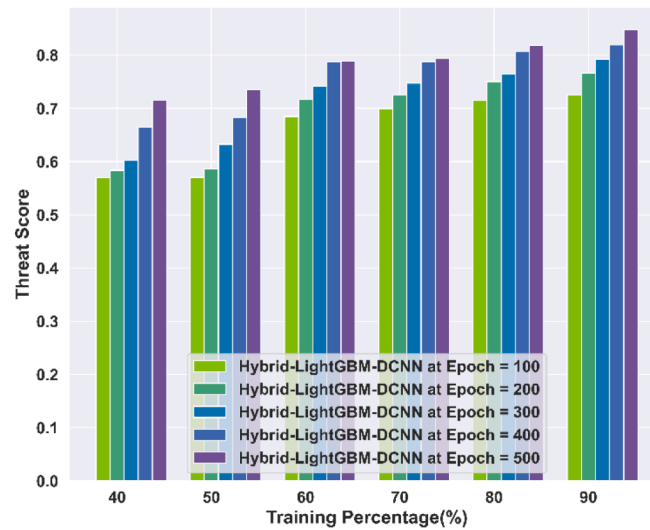
MCC



NPV



PPV



TS

Fig. 4. Performance analysis with TP.

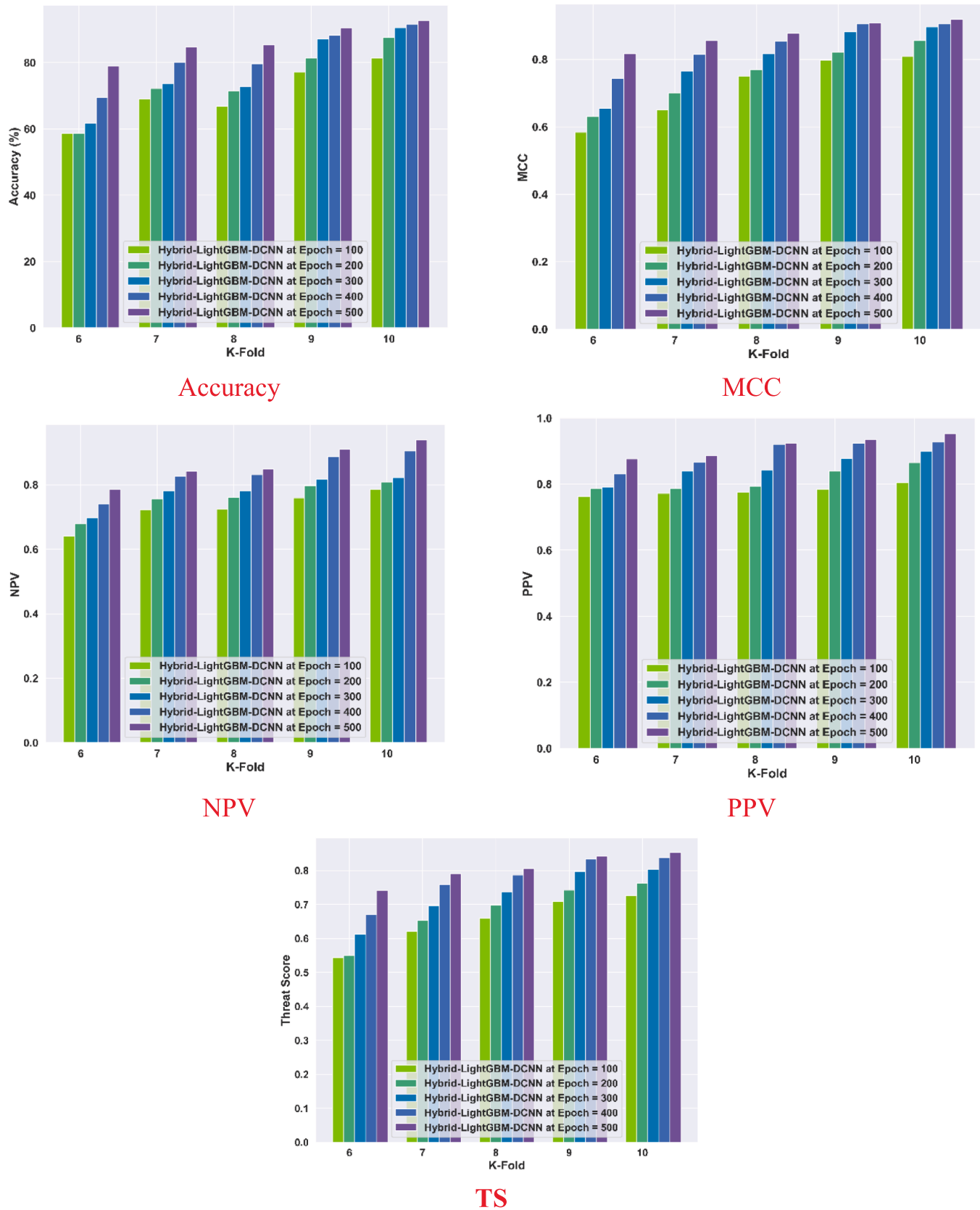


Fig. 5. Performance analysis with k-fold 10.

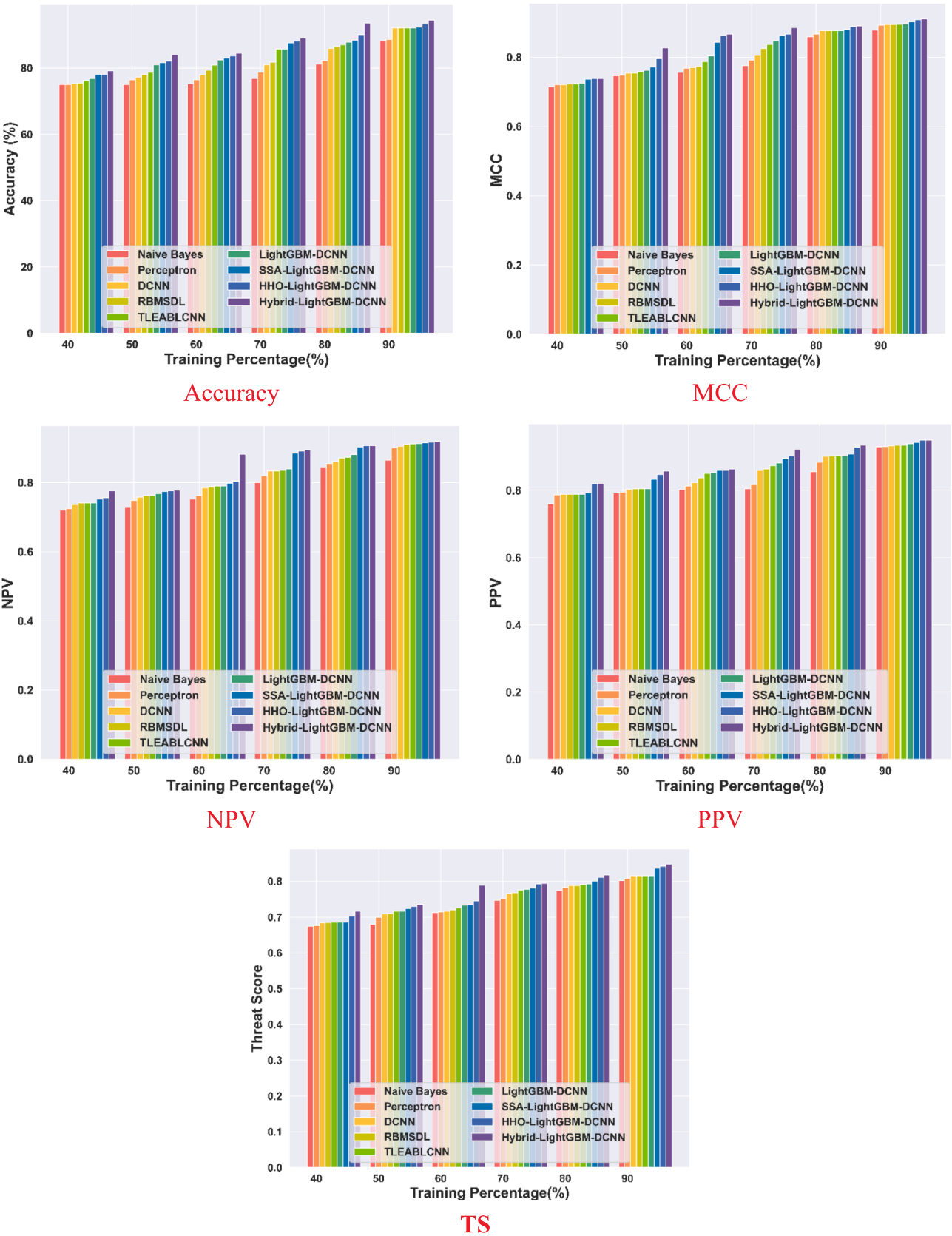


Fig. 6. Comparative analysis with TP.

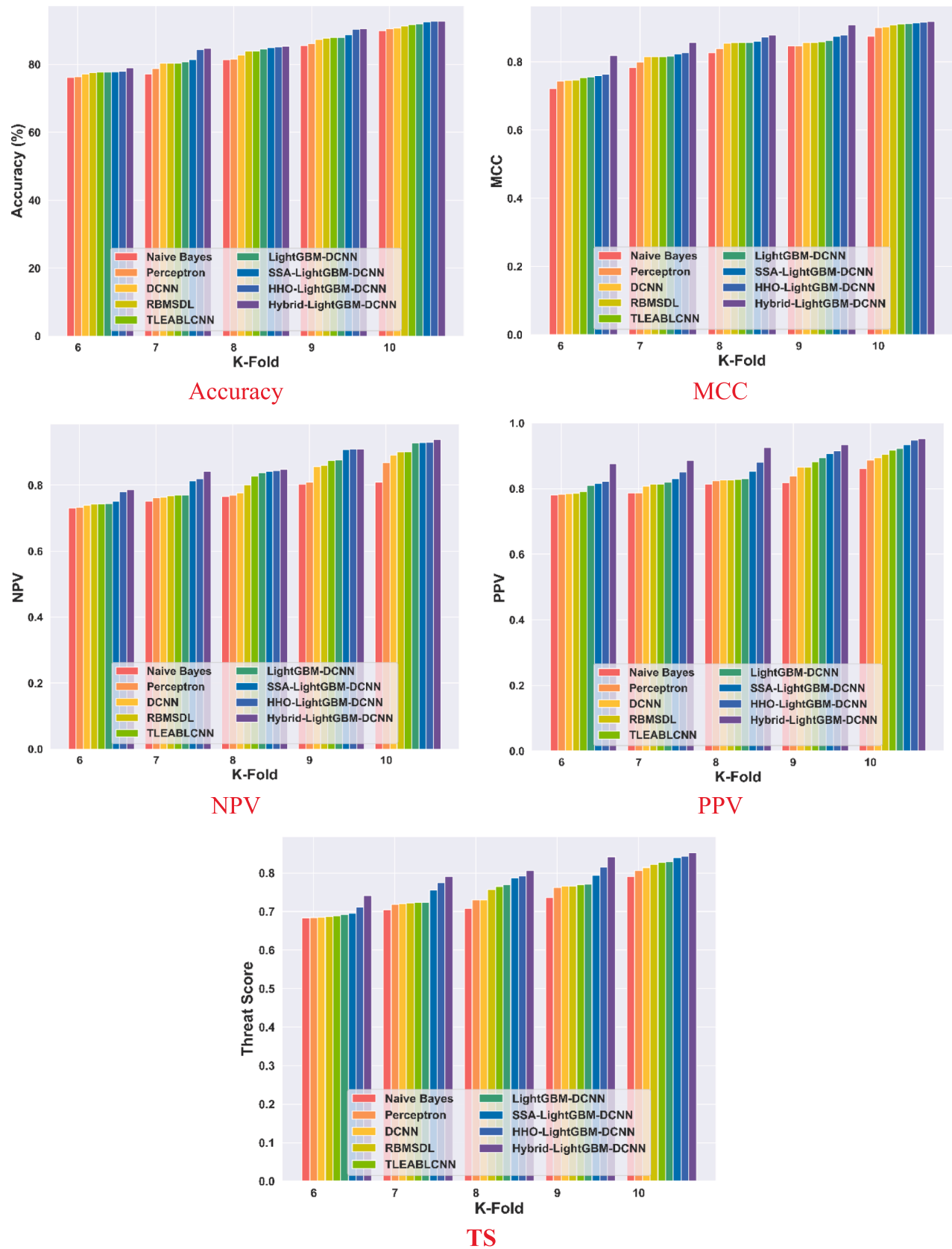


Fig. 7. Comparative analysis with k-fold.

consists of details of 819 people in the age groups of 55–90 years and above. Out of which 229 people were normal, 398 patients were affected by MCI, and 192 with AD. The main aim of the ADNI is to develop better methods for gaining MRI and PET data from patients with MCI, AD, and elderly people, to describe the structural changes in the brain, and also to enhance the AD detection methods for providing better treatment. In this research, the input data is equally assigned or balanced into batches. Further, the balanced data are split as 90 % for training and 10 % for testing purposes.

4.3. Experimental results

The experimental results for the PFO-hybrid light GBM-based deep CNN model's capability to identify AD are shown in Fig. 3. This graphical representation shows a range of images, such as the original input, segmentation images, feature extraction images, and the outputs are depicted below.

4.4. Evaluation metrics

The performance of the proposed model is evaluated based on the performance metrics such as accuracy, Matthew's correlation coefficient (MCC), Negative Predictive Value (NPV), Positive Predictive Value (PPV), and threat score, which evaluate the effectiveness of AD detection. Accuracy calculates the ratio of correctly predicted values to the total predictions. Matthew's correlation coefficient demonstrates the balanced measure of AD detection based on the true positives, true negatives, false positives, and false negatives. The NPV metric gives the probability of the negative test results in a patient, where the positive NPV value represents the Alzheimer's true negative, and the negative NPV denotes a false negative. The PPV metric explains the probability of the positive test results in a patient, where the positive PPV represents the true positive, and the negative NPV denotes a false positive. Threat score (TS) measures the ratio of true positive predictions with remaining all the positive predictions. The mathematical formulations of performance metrics are represented as follows.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (50)$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (51)$$

$$\text{NPV} = \frac{TN}{TN + FN} \quad (52)$$

$$\text{PPV} = \frac{TP}{TP + FP} \quad (53)$$

$$\text{TS} = \frac{TP}{TP + FN + FP} \quad (54)$$

4.5. Performance analysis

The Performance assessment of the proposed model is conducted using the ADNI dataset at multiple epochs, i.e., 100 intervals, encompassing epochs at 100, 200, 300, 400, and 500, to demonstrate the effectiveness of the recommended approach for AD detection. As part of these studies, the TP and k-fold metrics are assessed, making it possible to monitor the model's evolution and performance over time.

4.5.1. Performance analysis with training percentage

Fig. 4 displays the findings from the PFO-hybrid light GBM-based deep CNN model, which was developed for the diagnosis of AD. When TP accuracy is tested at 90 %, the PFO-hybrid light GBM-based deep CNN model obtains the following scores: 87.92 %, 91.79 %, 92.00 %, 92.35 %, and 94.48 %. Similarly, the model produces the following

results when TP's Mathew correlation coefficient is evaluated at 90 %: 86.49 %, 89.25 %, 89.80 %, 90.12 %, and 91.00 %. Furthermore, while evaluating the TP negative predictive value at 90 %, the PFO-hybrid light GBM-based deep CNN achieves the following values: 84.98 %, 87.22 %, 89.30 %, 90.19 %, and 91.83 %. Analyzing the positive predictive value graph for TP at the 90 % level, the model also yields the following results: 86.69 %, 90.73 %, 91.76 %, 92.37 %, and 94.97 %. Lastly, the model produces the following outcomes when TP's thread score graph is evaluated at 90 %: 79.22 %, 81.95 %, 83.26 %, 84.24 %, and 84.77 %.

4.5.2. Performance analysis with k-fold

Fig. 5 displays the findings from the PFO-hybrid light GBM-based deep CNN model, which was developed for the diagnosis of AD. When accuracy is tested at 10-fold, the PFO-hybrid light GBM-based deep CNN model obtains the following scores: 89.97 %, 90.89 %, 91.41 %, 91.73 %, and 92.70 %. Similarly, the model produces the following results when the Mathew correlation coefficient graph is evaluated at 10-fold: 88.25 %, 90.54 %, 90.66 %, 90.91 %, and 91.85 %. Furthermore, while evaluating the TP negative predictive value graph at 10-fold, the PFO-hybrid light GBM-based deep CNN achieves the following values: 82.27 %, 88.68 %, 90.45 %, 91.53 %, and 93.91 %. Analyzing the positive predictive value graph for 10-fold, the model also yields the following results: 89.86 %, 92.05 %, 92.32 %, 93.23 %, and 95.32 %. Lastly, the model produces the following outcomes when TP's thread score graph is evaluated at 10-fold: 82.33 %, 82.57 %, 84.64 %, 85.15 %, and 85.27 %.

4.6. Comparative methods

The comparative analysis of the proposed PFO-hybrid Light GBM-based DCNN model is evaluated based on training percentage and k-fold value using the ADNI dataset. The model is compared with other existing models, such as Naive Bayes [24], Perceptron [25], DCNN [26], Light GBM-DCNN [27], SSA-Light GBM-DCNN [28], HHO-Light GBM-DCNN [29], RBMSDL [30], and TLEABLCNN [31] to improve the performance for AD detection.

4.6.1. Comparative analysis with training percentage

Fig. 6 depicts the diagnostic performance for AD compared with other existing models, focusing specifically on Training Percentage. During evaluation, the proposed PFO-hybrid Light GBM-based DCNN model exhibits a noteworthy gain in 90 % of training. The proposed model achieved a high accuracy of 94.48 %, which is 2.46 % higher than the RBMSDL model. Additionally, the proposed model attained an MCC value of 0.91, which is 0.91 % more than the RBMSDL model. At 90 % training yields a NPV of 0.92 for the PFO-hybrid Light GBM-based DCNN model, which is 0.92 % higher than the RBMSDL model. Furthermore, the suggested PFO-hybrid Light GBM-based DCNN gained a PPV of 0.95, which is 1.65 % more compared with the RBMSDL model. Similarly, the PFO-hybrid Light GBM-based DCNN model shows a significant increase of 3.78 % over the RBMSDL model while evaluating the threat score for training at the 90 %.

4.6.2. Comparative analysis with k-fold

Fig. 7 depicts the diagnostic performance for AD compared with other existing models, focusing specifically on Training Percentage. During evaluation, the proposed PFO-hybrid Light GBM-based DCNN model exhibits a noteworthy gain in 10 k-fold value. The proposed model achieved a high accuracy of 92.7 %, which is 1.43 % higher than the RBMSDL model. Additionally, the proposed model attained an MCC value of 0.92, which is 1.12 % more than the RBMSDL model. And the model yields an NPV of 0.94 for the PFO-hybrid Light GBM-based DCNN model and compared to 4.1 % higher than the RBMSDL model. Furthermore, the suggested PFO-hybrid Light GBM-based DCNN gained a PPV of 0.95, which is 5.01 % more compared with the RBMSDL model.

Table 1

Comparative discussion of the proposed PFO-hybrid light GBM-based DCNN.

Methods	Training percentage-90 %					k-fold value-10				
	Accuracy (%)	MCC	NPV	PPV	TS	Accuracy (%)	MCC	NPV	PPV	TS
Naive Bayes	88.24	0.88	0.86	0.93	0.80	89.88	0.88	0.81	0.86	0.79
Perceptron	88.60	0.89	0.90	0.93	0.81	90.48	0.90	0.87	0.89	0.81
DCNN	92.13	0.90	0.90	0.93	0.82	90.79	0.90	0.89	0.89	0.81
RBMSDL	92.16	0.90	0.91	0.93	0.82	91.38	0.91	0.90	0.91	0.82
TLEABLCNN	92.16	0.90	0.91	0.94	0.82	91.70	0.91	0.90	0.92	0.83
Light GBM-DCNN	92.18	0.90	0.91	0.94	0.82	91.99	0.91	0.93	0.92	0.83
SSA-Light GBM-DCNN	92.39	0.90	0.91	0.94	0.84	92.47	0.91	0.93	0.93	0.84
HHO-Light GBM-DCNN	93.48	0.91	0.92	0.95	0.84	92.66	0.92	0.93	0.95	0.84
Hybrid-Light GBM-DCNN	94.48	0.91	0.92	0.95	0.85	92.70	0.92	0.94	0.95	0.85

Table 2

Statistical analysis.

Models			Naive Bayes	Perceptron	DCNN	RBMSDL	TLEABLCNN	Light GBM-DCNN	SSA-Light GBM-DCNN	HHO-Light GBM-DCNN	Hybrid-Light GBM-DCNN
Training Percentage-90 %	Accuracy	Best	88.24	88.60	92.13	92.16	92.16	92.18	92.39	93.48	94.48
		Mean	78.61	79.59	81.57	82.17	83.44	84.32	85.18	85.95	87.47
		Variance	23.33	21.46	33.89	31.65	29.46	24.55	22.71	26.38	29.89
		Standard Deviation	4.83	4.63	5.82	5.63	5.43	4.95	4.77	5.14	5.47
		MCC	0.88	0.89	0.90	0.90	0.90	0.90	0.90	0.91	0.91
	MCC	Best	0.88	0.89	0.90	0.90	0.90	0.90	0.90	0.91	0.91
		Mean	0.79	0.80	0.80	0.81	0.81	0.82	0.83	0.84	0.85
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
	NPV	Best	0.86	0.90	0.90	0.91	0.91	0.91	0.91	0.92	0.92
		Mean	0.78	0.80	0.81	0.82	0.82	0.82	0.84	0.84	0.86
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.07	0.06
	PPV	Best	0.93	0.93	0.93	0.93	0.94	0.94	0.94	0.95	0.95
		Mean	0.82	0.84	0.85	0.85	0.86	0.86	0.87	0.88	0.89
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
	Threat Score	Best	0.80	0.81	0.82	0.82	0.82	0.82	0.84	0.84	0.85
		Mean	0.73	0.74	0.75	0.75	0.75	0.75	0.76	0.77	0.78
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
K-fold 10	Accuracy	Best	89.88	90.48	90.79	91.38	91.70	91.99	92.47	92.66	92.70
		Mean	82.03	82.68	83.66	84.21	84.36	84.63	85.08	86.08	86.48
		Variance	26.63	25.63	23.73	24.58	24.89	25.16	26.62	26.27	23.02
		Standard Deviation	5.16	5.06	4.87	4.96	4.99	5.02	5.16	5.13	4.80
		MCC	0.88	0.90	0.90	0.91	0.91	0.91	0.91	0.92	0.92
	MCC	Best	0.88	0.90	0.90	0.91	0.91	0.91	0.91	0.92	0.92
		Mean	0.81	0.83	0.83	0.84	0.84	0.84	0.85	0.85	0.88
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.04
	NPV	Best	0.81	0.87	0.89	0.90	0.90	0.93	0.93	0.93	0.94
		Mean	0.77	0.79	0.81	0.81	0.82	0.83	0.85	0.86	0.86
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.03	0.05	0.06	0.06	0.06	0.07	0.06	0.06	0.05
	PPV	Best	0.86	0.89	0.89	0.91	0.92	0.92	0.93	0.95	0.95
		Mean	0.81	0.82	0.84	0.84	0.85	0.86	0.87	0.88	0.91
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.03	0.04	0.04	0.04	0.05	0.04	0.04	0.04	0.03
	Threat Score	Best	0.79	0.81	0.81	0.82	0.83	0.83	0.84	0.84	0.85
		Mean	0.72	0.74	0.74	0.75	0.76	0.76	0.77	0.79	0.81
		Variance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Standard Deviation	0.04	0.04	0.04	0.05	0.05	0.05	0.05	0.04	0.04

Similarly, the PFO-hybrid Light GBM-based DCNN model shows a significant increase of 3.62 % over the RBMSDL model while evaluating the threat score for a k-fold value of 10.

4.7. Comparative discussion

This section provides a detailed comparison study of the proposed PFO-hybrid Light GBM-based DCNN, with an emphasis on the

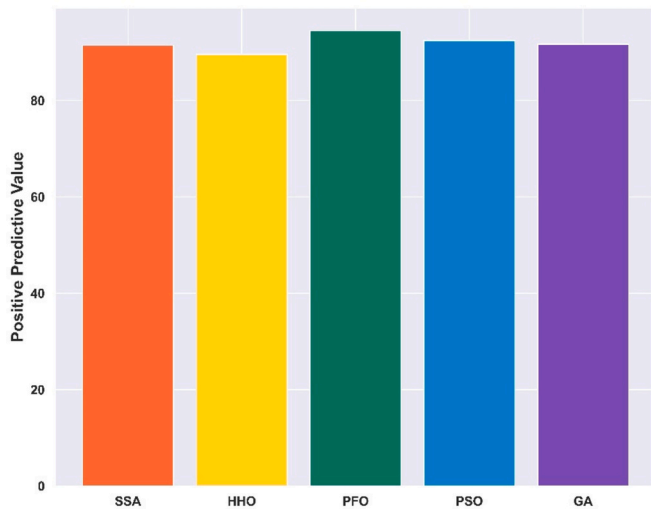


Fig. 8. Ablation study with multiple optimizations.

measurement of TP and k-fold. With its weights and bias carefully adjusted, the PFO-hybrid Light GBM-based DCNN shows impressive performance, resulting in a great model. The existing models have limitations such as complex structure, overfitting issues, computational complexity, and poor generalizability with limited resources. The proposed model effectively identifies Alzheimer's disease compared with other existing models. Further, the proposed PFO-hybrid light GBM-based CNN model is developed and evaluated using the ADNI dataset. Initially, preprocessing is performed to reduce inherent noise and artifacts. A Multi-level BFC Texture-Based Segmentation technique precisely separates brain tissues. Feature extraction includes statistical features, hybrid texture patterns, and VGG-16. The strengths of Hybrid Light GBM and CNN are combined for Alzheimer's detection. The PFO algorithm optimizes the segmentation process and the hybrid light GBM model, which helps to improve the performance and address overfitting issues. More specifically, the proposed model outperforms other existing models with good generalizability, and the simple with efficient structure of the model eliminates complexity and overfitting issues. The PFO-hybrid Light GBM-based DCNN outperformed other methods in the field, as shown in Table 1. To achieve fair and equitable assessments, this evaluation is conducted using a k-fold of 10 setup and 90 % TP.

4.8. Statistical analysis

The statistical analysis of the proposed PFO-hybrid Light GBM-based DCNN model is carried out to evaluate the robustness of the reported results. Further, the statistical measures, including the best, mean, variance, and Standard Deviation, are used in the statistical analysis for evaluating the statistical significance of the proposed model. The results obtained with the statistical analysis using the ADNI dataset are shown in Table 2 below.

4.9. Ablation study with multiple optimization

The proposed PFO algorithm is analyzed with other existing algorithms to enhance the performance of the algorithm based on the PPV value, which is plotted in Fig. 8. The ablation study was used to assess the functionality of the proposed model and demonstrates the significance of the PFO algorithm with existing algorithms. The evaluation results state that the proposed PFO algorithm attained a high PPV of 94.48 %, comparatively the other algorithm gained SSA of 91.47, HHO of 89.56, PSO of 92.44, and GA of 91.68, which are less compared with the proposed algorithm, which explains the superior performance of the PFO algorithm.

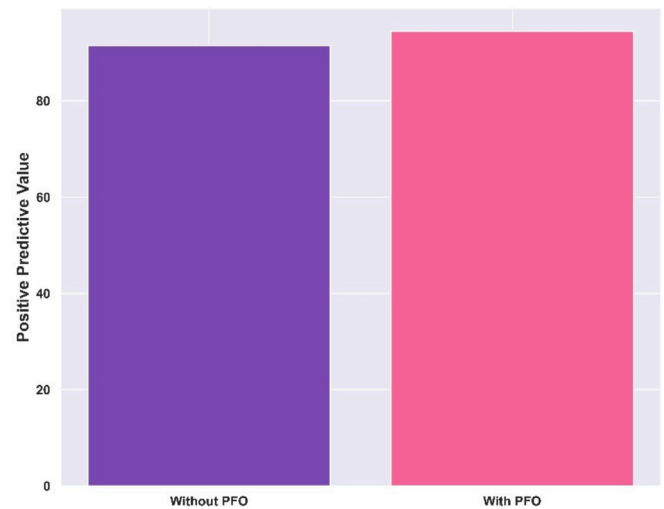


Fig. 9. Ablation study.

4.10. Ablation study

The proposed PFO-hybrid Light GBM-based DCNN model for AD detection utilizes the PFO algorithm for the segmentation and classification process, which enhances the accurate detection. The ablation study evaluates the accuracy of the model with the PFO algorithm and without the PFO algorithm based on accuracy, which enhances the detection performance of the model. The proposed model attained 94.48 % accuracy with the PFO algorithm and 91.46 % without the PFO algorithm, which is depicted in Fig. 9.

5. Conclusion

In this research, the proposed PFO-hybrid light GBM-based deep CNN model is developed for effective AD detection. The utilization of the Multi-level BFC Texture-Based Segmentation technique effectually separates affected brain tissues. The multiple feature extraction techniques enhance the accuracy of AD detection. Moreover, the integration of Hybrid Light GBM-based CNN with PFO algorithm fine-tunes the parameters of the model for accurate detection, which helps to improve the performance and address overfitting issues. The large collection of MRI images from ADNI datasets trains the model for unseen data and enhances the generalization, which is also valid for real-time applications. More specifically, the research evaluates the performance of the PFO-hybrid Light GBM-based DCNN model with existing models, achieving a high accuracy of 94.48 %, MCC of 0.91, NPV of 0.92, PPV of 0.95, and TS of 0.85 based on 90 % of training. The proposed PFO-hybrid Light GBM-based DCNN demonstrates an accuracy of 92.70 %, MCC of 0.92, NPV of 0.94, PPV of 0.95, and TS of 0.85 based on a 10-fold. Further, the proposed model will incorporate the explainability technique to augment the decision-making skill and be analyzed by image encryption techniques in future research work.

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Ethical and informed consent for the data used

Not applicable.

Data availability and access

The datasets used in this study are publicly available.

CRediT authorship contribution statement

Afnan M. Alhassan: Writing – original draft, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Nouf I. Altmami:** Writing – review & editing, Software, Resources, Project administration,

Methodology, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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